

## Lecture 8

### Orthonormal Bases, Gram–Schmidt, and Riesz

The core uses finite-dimensional inner-product spaces over  $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$  in the physics convention (conjugate-linear in the first slot). Tags mark the flavour; a ★ marks a harder one.

#### Core tutorial route

Work **8.2, 8.3, 8.5, 8.17, 8.8, 8.9, and 8.19** in that order (about 80–100 minutes before discussion). This seven-problem route covers coordinates and Parseval, real and complex Gram–Schmidt, an orthogonal complement/dimension check, best approximation, least-squares transfer, and a genuine finite Riesz explanation with the complex bra–ket convention.

**Additional practice:** 8.1, 8.4, 8.6–8.7, 8.10–8.14, 8.16, 8.18, 8.20, 8.24. **Bridge:** 8.15, 8.21–8.22 (integral/Frobenius models and a locally defined adjoint preview). **Extension:** 8.23 (codimension-one forward link). Extension tasks do not count toward core progress.

#### Orthonormal bases and Gram–Schmidt

**Exercise 8.1** (computational) Apply Gram–Schmidt to  $v_1 = (1, 1, 1)$ ,  $v_2 = (1, 1, 0)$ ,  $v_3 = (1, 0, 0)$  in  $\mathbb{R}^3$  with the dot product.

**Exercise 8.2** (computational) Expand  $v = (2, 3) \in \mathbb{R}^2$  in the orthonormal basis  $e_1 = \frac{1}{\sqrt{2}}(1, 1)$ ,  $e_2 = \frac{1}{\sqrt{2}}(1, -1)$ , and verify Parseval’s relation.

**Exercise 8.3** (computational) Apply Gram–Schmidt to  $v_1 = (3, 4)$ ,  $v_2 = (1, 0)$  in  $\mathbb{R}^2$  with the dot product.

**Exercise 8.4** (computational) Expand  $v = (3, 0, 0) \in \mathbb{R}^3$  in the orthonormal basis  $e_1 = \frac{1}{3}(1, 2, 2)$ ,  $e_2 = \frac{1}{3}(2, 1, -2)$ ,  $e_3 = \frac{1}{3}(2, -2, 1)$ , and check Parseval’s relation.

**Exercise 8.5** (computational) Orthonormalize  $v_1 = (1, i)$ ,  $v_2 = (0, 1)$  in  $\mathbb{C}^2$  (physics convention, conjugate-linear in the first slot).

**Exercise 8.6** (proof) Prove that an orthonormal list  $e_1, \dots, e_m$  is linearly independent directly from the definition.

**Exercise 8.7** (proof) Let  $e_1, \dots, e_n$  be an orthonormal basis of  $V$ . Prove that for all  $u, v \in V$ ,

$$\langle u, v \rangle = \sum_{k=1}^n \overline{\langle e_k, u \rangle} \langle e_k, v \rangle,$$

so the coordinate map  $v \mapsto (\langle e_1, v \rangle, \dots, \langle e_n, v \rangle)$  preserves inner products.

#### Projections and best approximation

**Exercise 8.8** (computational) Find the point of the line  $U = \text{span}((1, 2, 2)) \subset \mathbb{R}^3$  closest to  $b = (3, 0, 0)$ .

**Exercise 8.9** (computational) Fit a line  $y = c_0 + c_1x$  to the data  $(1, 1), (2, 2), (3, 2)$  by least squares.

**Exercise 8.10** (computational) Find the orthogonal projection of  $b = (2, 4) \in \mathbb{R}^2$  onto the line  $U = \text{span}((1, 1))$ , and the distance from  $b$  to  $U$ .

**Exercise 8.11** (computational) Project  $b = (3, 0, 0) \in \mathbb{R}^3$  onto the plane  $U$  with orthonormal basis  $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ ,  $e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$ .

**Exercise 8.12** (computational) Solve the inconsistent system  $Ax = b$  in the least-squares sense, where  $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$  and  $b = (1, 1, 0)$ .

**Exercise 8.13** (proof) Suppose  $P = P_U$  is an orthogonal projection. Prove  $P^2 = P$  and that  $v - Pv \perp Pv$  for every  $v$ .

**Exercise 8.14** (proof) Let  $P_U$  be the orthogonal projection onto a subspace  $U$  with orthonormal basis  $e_1, \dots, e_m$ . Prove that  $\langle P_U v, w \rangle = \langle v, P_U w \rangle$  for all  $v, w \in V$ .

**Exercise 8.15** (★) On  $\mathcal{P}_2$  with  $\langle p, q \rangle = \int_{-1}^1 \bar{p} q \, dx$ , find the polynomial of degree  $\leq 1$  closest to  $f(x) = x^2$ , and verify that the error is orthogonal to  $\mathcal{P}_1$ .

### Riesz and orthogonal complements

**Exercise 8.16** (computational) Find the vector  $w \in \mathbb{C}^2$  representing the functional  $\varphi(v) = 2v_1 - iv_2$  in the form  $\varphi(v) = \langle w, v \rangle$ .

**Exercise 8.17** (computational) For  $U = \text{span}((1, 0, 1)) \subset \mathbb{R}^3$ , describe  $U^\perp$  and verify  $\dim U + \dim U^\perp = 3$ .

**Exercise 8.18** (computational) Find  $w \in \mathbb{R}^3$  representing the functional  $\varphi(v) = 2v_1 - v_2 + 3v_3$  in the form  $\varphi(v) = \langle w, v \rangle$ .

**Exercise 8.19** (proof) State and prove the finite-dimensional Riesz representation theorem for a linear functional  $\varphi: V \rightarrow \mathbb{F}$ , including its hypotheses, existence, and uniqueness. Then explain why  $w \mapsto \langle w, \cdot \rangle$  is conjugate-linear over  $\mathbb{C}$  but linear over  $\mathbb{R}$ .

**Exercise 8.20** (computational) For  $U = \text{span}((1, 0, 1, 0), (0, 1, 0, 1)) \subset \mathbb{R}^4$ , give a basis of  $U^\perp$  and check  $\dim U + \dim U^\perp = 4$ .

**Exercise 8.21** (computational) On  $\mathcal{M}_2(\mathbb{R})$  with the Frobenius product  $\langle A, B \rangle = \text{tr}(A^\dagger B)$ , find the matrix  $w$  representing  $\varphi(A) = 2A_{11} - A_{12} + A_{22}$  as  $\varphi(A) = \langle w, A \rangle$ .

**Exercise 8.22** (★) **Forward bridge.** For a matrix/operator  $T$  in an ONB, define its adjoint  $T^\dagger$  by  $\langle T^\dagger u, v \rangle = \langle u, Tv \rangle$  (equivalently, conjugate transpose of the matrix). Prove that  $\text{tr}(T^\dagger T) = \sum_{j,k} |\langle e_j, Te_k \rangle|^2 \geq 0$ , with equality iff  $T = 0$ .

**Exercise 8.23** (★) Suppose  $V$  is finite-dimensional and  $\varphi$  is a linear functional on  $V$  represented by  $w$  via  $\varphi(v) = \langle w, v \rangle$ . Prove that  $\text{null } \varphi = (\text{span}(w))^\perp$  and, if  $w \neq 0$ , that  $\dim \text{null } \varphi = \dim V - 1$ .

**Exercise 8.24** (proof) Let  $U$  be a subspace of an inner product space  $V$ . Prove directly from the definitions that  $U \subseteq (U^\perp)^\perp$ .